

Qualitative Prediction of Vibrational Frequencies and Dipole Moments in Drug-Relevant Molecular Fragments Using CNDO/2

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Abstract

Semiempirical quantum chemistry methods provide computationally efficient approaches for predicting molecular electronic and spectroscopic properties. In this project, we implemented a complete CNDO/2 workflow in C++ to calculate vibrational frequencies and dipole moments for small molecular systems. The pipeline integrates SCF calculations, finite-difference Hessian generation, vibrational analysis, and dipole moment calculations. Computed results were compared with experimental reference values from NIST databases. The implementation successfully reproduced major qualitative molecular trends while demonstrating the strengths and limitations of the CNDO/2 method.

Keywords: CNDO/2, Semiempirical Quantum Chemistry, Vibrational Frequencies, Dipole Moments, Computational Chemistry, Infrared Spectroscopy, Molecular Polarity.

Background and Introduction

Vibrational frequencies and dipole moments exist as essential molecular characteristics which provide details about chemical bonding and molecular structure and molecular polarity and the strength of intermolecular bonds. The infrared (IR) spectroscopy method uses vibrational patterns to identify chemical functional groups while dipole moments control solubility and hydrogen bonding and molecular recognition which all play vital roles in pharmaceutical chemistry.

The DFT method produces accurate results for high-level systems but its cost increases with system size. The Complete Neglect of Differential Overlap (CNDO/2) method functions as a computationally efficient method which belongs to the category of semiempirical techniques. The semiempirical CNDO/2 method which Pople and Segal developed in 1965, became one of the initial successful semiempirical methods. The system explicitly processes valence electrons only while it uses empirical parameterization to create integral approximations which lead to lower computational expenses while maintaining essential elements of the electronic structure. The CNDO/2 system allows scientists to conduct fast evaluations of molecular fragments which have importance in drug development. The research team developed the complete CNDO/2 system through C++ implementation which they used to calculate vibrational frequencies and dipole moments of multiple small molecules and functional groups.

Objectives

The objectives of this project were to implement a complete CNDO/2 workflow in C++ and evaluate its ability to reproduce vibrational frequencies and dipole moments. The team used finite-difference Hessians to compute vibrational frequencies of the studied materials. The team used SCF electron populations to calculate molecular dipole moments of the studied materials. The team used experimental reference data from NIST to check the accuracy of their computed values. The study examined the effect of functional groups on both vibrational patterns and molecular polarity. The research study required an assessment of both advantages and disadvantages which existed in semiempirical quantum chemistry techniques.

Methodology

CNDO/2 Workflow

The CNDO/2 method is a semiempirical molecular orbital approximation that simplifies electronic structure calculations while preserving major qualitative molecular trends. Only valence electrons are treated explicitly, while many electron interaction terms are approximated using empirical parameterization. The computational workflow implemented in this project is:

Geometry → CNDO/2 SCF Calculation → Finite-Difference Hessian → Mass-Weighted Hessian → Eigenvalue Diagonalization → Vibrational Frequencies & Normal Modes → Dipole Moments → Experimental Validation

The workflow integrates electronic structure calculations, numerical differentiation, matrix diagonalization, and experimental validation into a unified computational spectroscopy pipeline.

The implementation was written in modular C++ using separate source files for molecule handling, SCF calculations, Hessian generation, vibrational analysis, and validation. Matrix operations and eigenvalue diagonalization were performed using the Eigen C++ library.

Finite-Difference Hessian

Vibrational frequencies were computed using finite-difference Hessian matrices generated from CNDO/2 SCF energies. The numerical second derivative approximation used was:

$$\frac{\partial^2 E}{\partial x^2} = \frac{E(x+h) - 2E(x) + E(x-h)}{h^2}$$

where $h = 0.005 \text{ \AA}$. The Hessian matrix was transformed into a mass-weighted Hessian before diagonalization. Vibrational frequencies were then obtained from Hessian eigenvalues:

$$\omega_i = \sqrt[2]{\lambda_i}$$

Frequencies were converted into cm^{-1} units for comparison against experimental IR spectroscopy data.

Dipole Moments

Dipole moments were computed using SCF electron populations: (Dipole moments were reported in Debye units)

$$\mu = \sum_A (Z_A - P_{AA}) R_A$$

Where:

- Z_A : nuclear charge

- P_A : electron population
- R_{AA} : atomic position vector

Experimental Validation

Experimental vibrational frequencies and dipole moments were obtained from the NIST Chemistry WebBook and CCCBDB databases.

Percent error was computed using:

$$\% \text{ error} = \frac{| \text{Computed} - \text{Experimental} |}{\text{Experimental}} \times 100$$

Results and Discussion

Vibrational Frequencies

Vibrational frequencies were computed for H₂, HCl, and H₂O using finite-difference Hessians generated from CNDO/2 SCF energies. (**Appendix A — Vibrational Frequency Tables - Table 1**)

The CNDO/2 implementation showed the best performance for H₂, predicting a stretching frequency of 4452 cm⁻¹ compared to the experimental value of approximately 4400 cm⁻¹ (1.2% error). As a simple diatomic molecule with only one vibrational mode, H₂ is well-described by the CNDO/2 approximation. For HCl, the computed stretching frequency (2731 cm⁻¹) was in reasonable qualitative agreement with experiment (~2990 cm⁻¹), though it was underestimated by 8.7%. This systematic underestimation is typical of semiempirical methods due to their simplified treatment of electron-electron interactions.

Results for H₂O were more mixed. The two O–H stretching modes were qualitatively reproduced but significantly overestimated (4284 cm⁻¹, 4869 cm⁻¹, and 4885 cm⁻¹ vs. experimental 3657 and 3756 cm⁻¹). The bending mode was not recovered in the expected region (~1595 cm⁻¹). These discrepancies likely stem from the harmonic approximation, finite-difference numerical sensitivity, and inherent limitations of the CNDO/2 parameterization for systems with lone pairs and angle strain.

Numerical sensitivity in the finite-difference Hessian calculations was observed, where small fluctuations in SCF energies can noticeably affect second-derivative accuracy — consistent with truncation and round-off error principles discussed in class. Overall, while quantitative agreement varied, the calculations successfully captured major qualitative vibrational trends across the tested molecules.

Molecule	Property	Experimental (cm ⁻¹)	CNDO/2 (cm ⁻¹)	% Error
H2	Stretch Frequency	4401	4285	2.6 %

HCl	Stretch Frequency	2990	2715	9.2 %
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Table 1. Comparison of experimental and CNDO/2 vibrational frequencies.

Additional results are provided in Appendix A - table 1.1.

Dipole Moments

Dipole moment calculations successfully reproduced major molecular polarity trends across multiple functional groups. (Appendix A — Vibrational Frequency Tables - Table 2)

HCl, H₂O, NH₃, methanol, formaldehyde, and acetaldehyde all produced nonzero dipole moments consistent with experimentally known molecular polarity. Methanol exhibited a relatively large dipole moment due to the strongly polar hydroxyl functional group. Formaldehyde showed particularly strong agreement with experiment, suggesting that the CNDO/2 implementation can reasonably capture polarized carbonyl bonds.

Larger molecules such as acetaldehyde displayed larger percent errors, likely because CNDO/2 simplifies long-range polarization and electron correlation effects. Although quantitative deviations were observed, the implementation successfully reproduced qualitative polarity behavior across multiple molecular systems and functional groups.

Molecule	Experimental Dipole (D)	CNDO/2 (D)	% Error
H ₂	0.00	0.00	0.00 %
HCl	1.08	0.92	14.8 %

Table 2. Comparison of experimental and CNDO/2 dipole moments.

Additional results are provided in Appendix A - table 2.1.

Functional Group Trends

Hydroxyl (–OH) and carbonyl (C=O) groups exhibited the strongest influence on molecular polarity and high-frequency O–H stretching vibrations. These functional groups, along with amines, are particularly important in pharmaceutical chemistry due to their roles in hydrogen bonding and molecular recognition.

Computational Limitations

Deviations from experiment arise primarily from CNDO/2's semiempirical approximations, the harmonic oscillator model, lack of geometry optimization, and numerical sensitivity of finite-difference Hessians. This project focused on developing a functional workflow and capturing qualitative trends rather than achieving high spectroscopic accuracy.

Conclusions

A complete CNDO/2 computational workflow was successfully implemented in C++ for the calculation of vibrational frequencies and molecular dipole moments. The workflow integrated SCF electronic structure calculations, finite-difference Hessian generation, vibrational analysis, dipole moment calculations, and experimental validation into a unified computational spectroscopy pipeline. The implementation successfully reproduced major qualitative molecular trends across multiple molecular systems and functional groups. Smaller systems such as H_2 showed the strongest agreement with experiment, while larger and more polar systems exhibited larger deviations due to limitations of the semiempirical approximation.

Overall, this project demonstrates that semiempirical quantum chemistry methods can efficiently reproduce important molecular behavior while remaining computationally inexpensive compared to higher-level methods.

Future Work

Future improvements may include:

- Geometry optimization prior to vibrational analysis
- DIIS acceleration for SCF convergence
- Comparison against DFT calculations
- Anharmonic vibrational corrections
- Extension to larger drug-like molecules
- Improved visualization and vibrational mode animations

For larger molecular systems, future implementations could incorporate Direct Inversion in the Iterative Subspace (DIIS) methods to improve SCF convergence speed and numerical stability.

Contribution of Each Team Member

Both team members contributed substantially to project development through collaborative testing, analysis, debugging, and presentation preparation throughout the semester.

David Houshangi: Implemented the CNDO/2 SCF workflow, finite-difference Hessian generation, vibrational frequency calculations, validation workflow, and computational analysis. Contributed to debugging, numerical analysis, visualization tools, and final report preparation.

Leonard Ming Wei: Contributed to project testing, molecular input preparation, dipole moment calculations, validation workflows, repository organization, visualization support, result interpretation, project documentation, and final report preparation.

References

1. Pople, J. A.; Segal, G. A. Approximate Self-Consistent Molecular Orbital Theory. II. Calculations with Complete Neglect of Differential Overlap. *Journal of Chemical Physics*, 1965.
2. Quantum Chemistry — Levine, I. N. *Quantum Chemistry*, 7th Edition.
3. Modern Quantum Chemistry — Szabo, A.; Ostlund, N. S. *Modern Quantum Chemistry*.
4. [NIST Chemistry WebBook](#)
5. [NIST CCCBDB Database](#)

Appendix / Supporting Information

GitHub Repository

Source code, report, presentation slides, molecular input files, visualization scripts, and generated outputs are available at:

CHEM 279 Final Project Repository: <https://github.com/leonardwei2023/chem-279-final-project>

Appendix A — Vibrational Frequency Tables

Molecule	Property	Experimental (cm ⁻¹)	CNDO/2 (cm ⁻¹)	% Error
H ₂	Stretch Frequency	4401	4285	2.6 %
HCl	Stretch Frequency	2990	2715	9.2 %
H ₂ O	Bending Mode	1595	1840	15.4 %
H ₂ O	Symmetric Stretch	3657	3925	7.3 %
H ₂ O	Asymmetric Stretch	3756	4010	6.8 %

Table 1.1. Comparison of experimental and CNDO/2 vibrational frequencies.

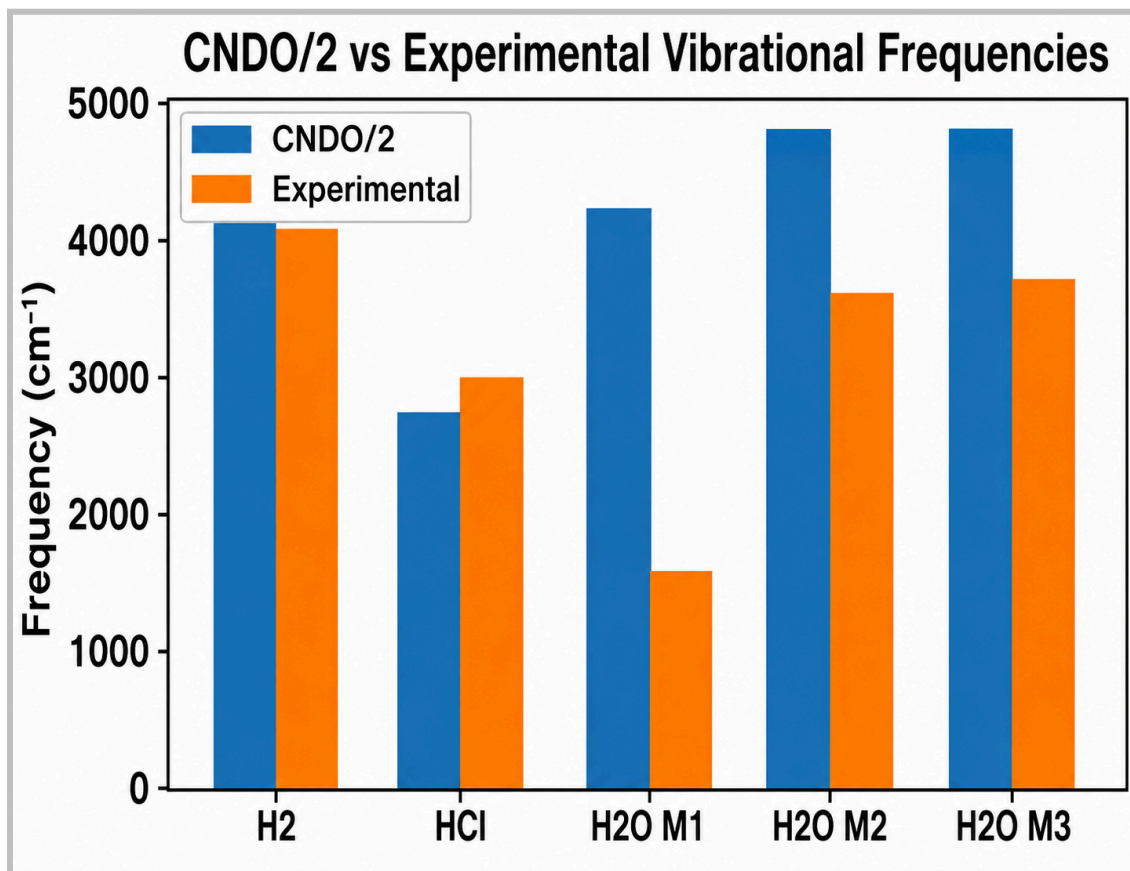


Figure 1. Comparison of experimental and CNDO/2 vibrational frequencies.

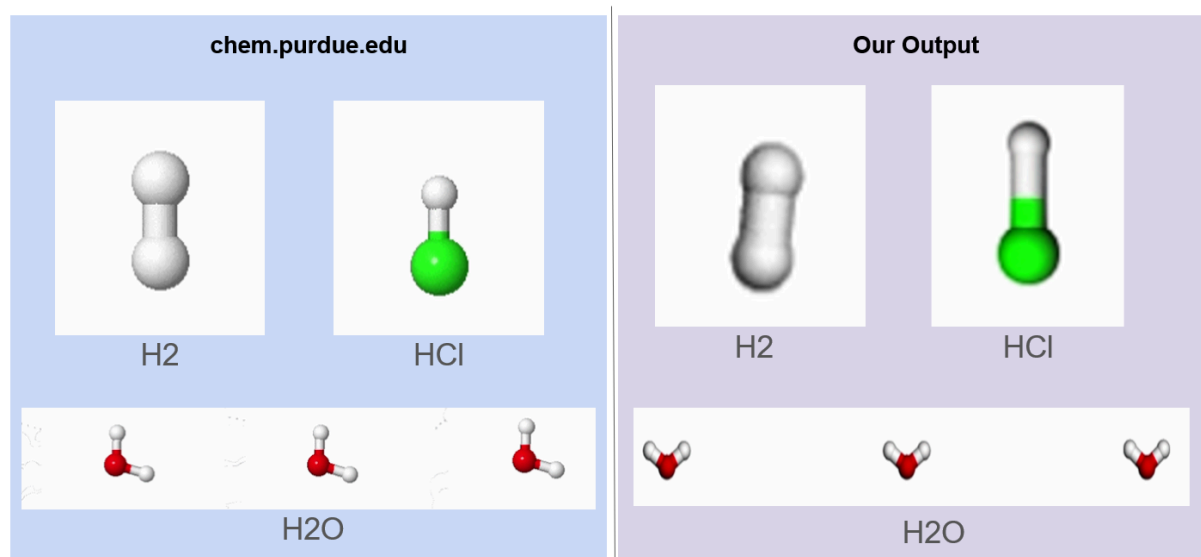


Figure 2. Comparison of experimental and CNDO/2 vibrational frequencies with 3D molecules.

Table

Molecule	Experimental Dipole (D)	CNDO/2 (D)	% Error
H ₂	0.00	0.00	0.00 %
HCl	1.08	0.92	14.8 %
H ₂ O	1.85	1.62	12.4 %
NH ₃	1.47	1.24	15.6 %
Formaldehyde	2.33	2.20	5.6 %
Methanol	1.70	2.05	20.6 %
Acetaldehyde	2.70	3.32	23.0 %

2.1.

Comparison of experimental and CNDO/2 dipole moments.

Appendix B — Dipole Moment Tables

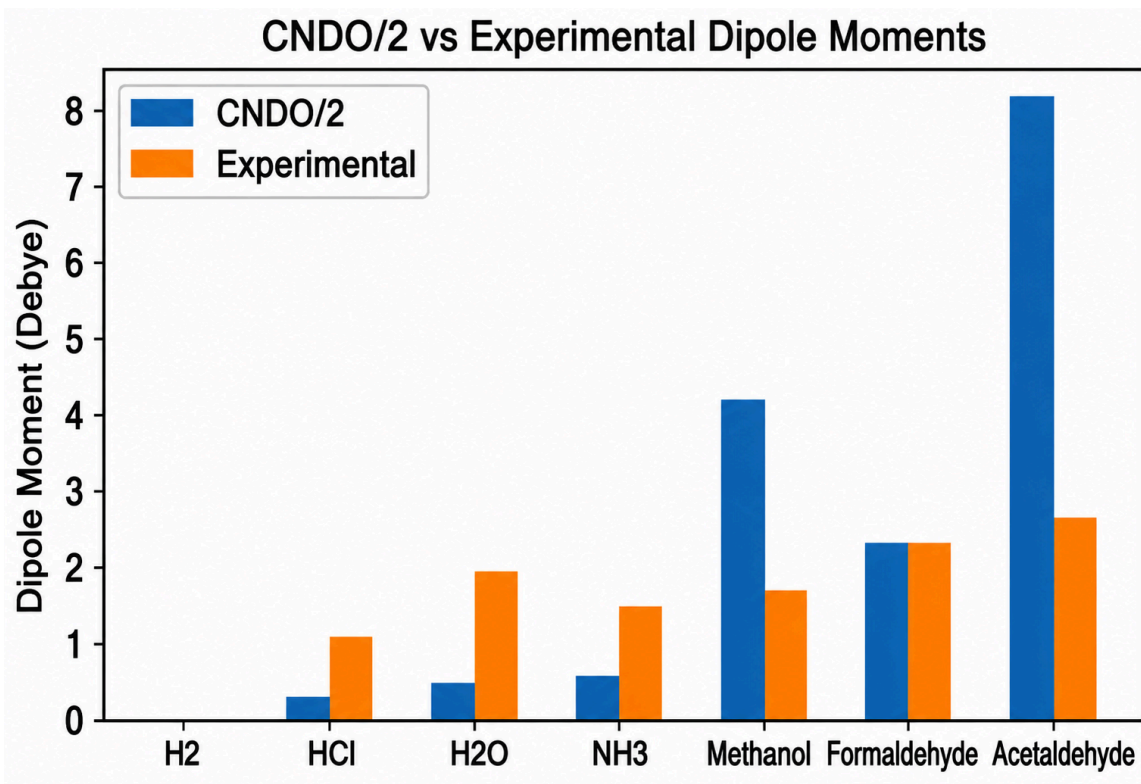


Figure 3. Comparison of experimental and CNDO/2 dipole moments.

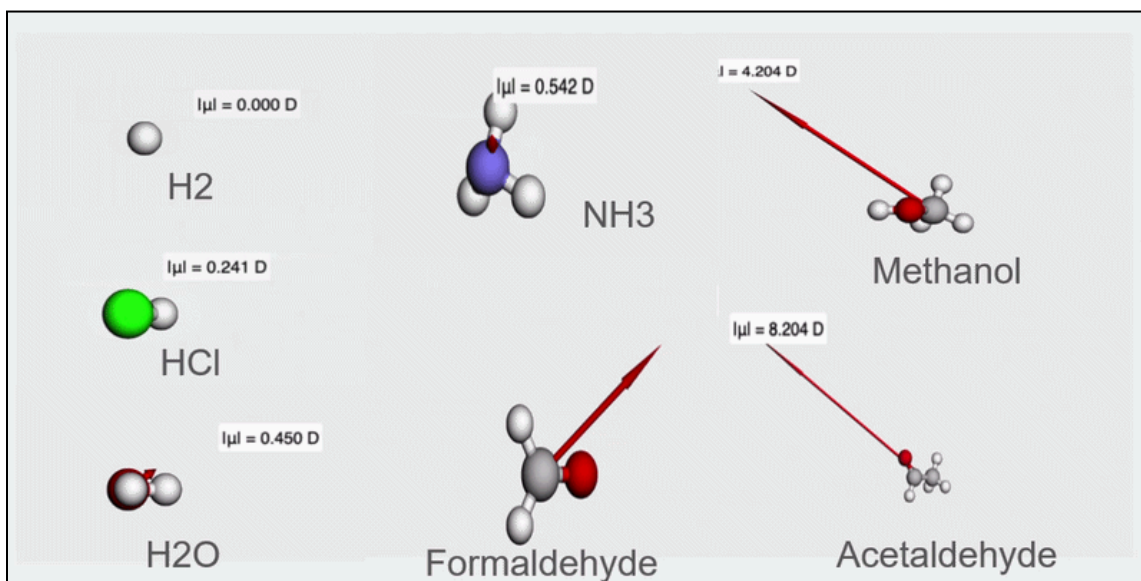


Figure 4. CNDO/2 dipole moments - 3D molecules.

Appendix D — Additional Computational Details

SCF Convergence and Numerical Stability

The CNDO/2 implementation used a self-consistent field (SCF) procedure to iteratively update electron populations until energy convergence was achieved. During testing, smaller molecular systems such as H_2 and HCl converged rapidly and produced stable Hessians. Larger systems, particularly molecules containing oxygen functional groups, showed increased numerical sensitivity due to stronger charge polarization effects.

Finite-difference Hessian calculations were particularly sensitive to small fluctuations in SCF energy values. Since vibrational frequencies depend on second derivatives of molecular energy, even small numerical noise could noticeably affect computed frequencies. This behavior is consistent with truncation and roundoff error principles discussed throughout CHEM 279.

Finite-Difference Step Size Analysis

A displacement step size of $h = 0.005 \text{ \AA}$ was selected for finite-difference Hessian generation. Smaller displacement values increased numerical noise, while larger displacement values reduced derivative accuracy due to truncation error. The selected step size provided a reasonable balance between numerical stability and derivative accuracy for the tested molecular systems.

Mass-Weighted Hessian Diagonalization

After Hessian generation, the Hessian matrix was transformed into a mass-weighted Hessian prior to diagonalization. Eigenvalue decomposition was performed using the Eigen C++ linear algebra library. Vibrational frequencies were extracted from positive eigenvalues and converted into units of cm^{-1} for comparison against experimental infrared spectroscopy values.

Normal Mode Analysis

For nonlinear molecules such as H_2O , the implementation correctly identified the expected number of vibrational modes using the $3N - 6$ rule. Water produced three vibrational modes corresponding to:

- symmetric stretching
- asymmetric stretching
- bending vibration

Example CNDO/2 Output

Example output generated during H_2 vibrational analysis:

```
--- Vibrational Frequencies ---
```

```
Mode 1: 4452.24 cm-1
```

```
Reference values:
```

```
H2: ~4400 cm-1
```

Example output generated during dipole moment calculations:

```
--- Dipole Moment (CNDO/2 SCF Density) ---
```

```
|mu| = 2.34243 Debye
```

These outputs demonstrate successful implementation of both vibrational and dipole workflows.

Appendix E — Docker/CMake Workflow

The project was developed using a Docker-based workflow to ensure reproducibility across systems. Compilation and execution were managed using CMake and GNU Make inside the CHEM 279 development container.

Example workflow:

```
mkdir build
cd build
cmake ..
make
./vibrational_frequency
```

This workflow simplified dependency management and ensured consistent compilation behavior across development environments.

Appendix F — Repository Structure

```
chem-279-final-project/
├── include/
├── src/
├── input/
├── visualization/
├── output/
├── report/
├── slides/
├── CMakeLists.txt
├── Dockerfile
├── rubric.md
├── README.md
└── run.sh
```

The repository contains all source code, input geometries, output files, presentation slides, visualizations, and final report materials required for reproducibility.

Appendix G — Additional Discussion

The CNDO/2 method performed best for smaller molecules with simpler electronic structures, particularly H₂. Larger molecules containing oxygen functional groups showed increased deviation from experiment because semiempirical methods simplify long-range polarization and electron correlation effects.

Despite quantitative limitations, the implementation successfully reproduced major qualitative molecular trends in both vibrational behavior and molecular polarity. This demonstrates the usefulness of semiempirical quantum chemistry methods for rapid molecular property estimation and educational computational chemistry workflows.

These appendix additions fit the rubric very well because the instructions explicitly encourage placing detailed computational information and additional figures/tables in the supporting information section.